PREETHAM A K

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PROFESSIONAL SUMMARY

Results-driven Cybersecurity Professional with a strong foundation in Information Security, Vulnerability Assessment, Risk Assessment, and Security Operations. Skilled in identifying and assessing security vulnerabilities, analyzing security threats, and evaluating security risks to help organizations strengthen their security posture. Combines technical problem-solving abilities with hands-on cybersecurity experience to support vulnerability management, security controls, threat mitigation, and continuous security improvement. Strong understanding of cybersecurity principles, security best practices, and risk management.

TECHNICAL SKILLS

Security Domains: SIEM Monitoring, Log Analysis, Incident Response, Threat Detection, Vulnerability Assessment, Malware Analysis, Phishing Analysis, Network Security, SOAR Fundamentals, IOC Analysis

Tools & Platforms: Wazuh, Splunk, Suricata, Nmap, Nessus, Wireshark, Zeek, VirusTotal

Programming: Python

Soft Skills: Communication, Critical Thinking, Analytical Problem-Solving

EDUCATION

JAIN (Deemed-to-be University), Bangalore, Karnataka *2024 – 2026*

Master of Computer Applications - Specialization: Information Security and Management System

CGPA: 8.71/10

BGS First Grade College, Mysore, Karnataka *2021 – 2024*

Bachelor of Computer Application(BCA)

CGPA: 9.25/10

PROJECTS

Cloud-Based Honeypot SOC Platform for Threat Detection and Security Monitoring *2025*

- Built a cloud-based honeypot SOC platform to attract, capture, and log real-world attack traffic for security monitoring and threat research.
- Analyzed captured attack data to identify common attacker tactics, source patterns, and exploitation attempts, translating raw logs into actionable threat intelligence.
- Developed a centralized dashboard for real-time visualization of security events, attacker geolocation, log analysis, and threat intelligence feeds.
- Correlated honeypot activity with known threat indicators to support proactive detection and improve situational awareness of ongoing attack trends.

Splunk Enterprise Setup and Data Ingestion *2025*

- Set up and configured Splunk Enterprise for log analysis, monitoring, and security event detection, including custom indexes for efficient, targeted searching.
- Ingested data via ZIP uploads and deployed the Splunk Universal Forwarder to collect and forward logs from remote machines for centralized analysis.
- Built search queries and basic correlation logic to identify anomalous activity within ingested log data, supporting threat identification and response workflows.
- Documented the ingestion pipeline and indexing structure to make the environment reproducible and easy to extend with additional log sources.

Wazuh Installation and Agent Deployment Lab *2025*

- Installed and configured Wazuh for centralized security monitoring, deploying agents across endpoints to improve threat visibility and detection coverage.
- Integrated the Wazuh Agent with the Wazuh Server by generating and executing agent installation and enrollment commands across multiple endpoints.
- Tuned Wazuh configuration and rule sets to reduce noise and surface higher-fidelity security alerts for triage.
- Currently expanding the lab environment further; full multi-endpoint deployment is in progress due to hardware resource constraints on the host system.

CERTIFICATIONS

- **Certified Cybersecurity Educator Professional | Red Team Leaders** *2026*
- **CompTIA Security+ (SY0-701) | CompTIA** *2026*
- **Endpoint Detection and Response (EDR) Foundation | Qualys** *2026*
- **Google Cybersecurity Professional Certificate | Coursera** *2025*
- **Security Operations Professional | Palo Alto Networks** *2025*
- **Intro to Splunk | Splunk** *2025*

PROFESSIONAL EXPERIENCE

Cybersecurity Intern — Redynox *Jun 2025 – Jul 2025*

- Performed vulnerability assessments and web application security testing using Nmap and Nessus; prioritized findings by severity and documented remediation recommendations.
- Monitored security events and analyzed logs on a daily basis, assisting in alert triage and contributing to incident investigation and response efforts.

PUBLICATIONS

A Multi-Layered Framework for DDoS Detection and Mitigation *2025*

- Proposed a three-layer defense architecture combining an application firewall, machine learning-based anomaly detection, and honeypot-integrated load balancing to counter DDoS attacks.
- Analyzed limitations of standalone DDoS mitigation techniques (firewalls, CAPTCHAs, rate limiting) and demonstrated how layered, adaptive defenses improve detection accuracy and reduce system downtime — presented at Jain University.

ACHIEVEMENTS

1st Place - “TRIVIDHA 2K24” Intercollegiate Technical Event *2024*

- Ranked 1st among competing teams at Trinity College, Mysore, in a technical event testing applied cybersecurity and problem-solving skills.